IM101
Finals Task #4
Jan Andrew Tobias
BSIT 3B

1. SQL statements (For creating the Student, assignment and submission (including the table relationship - using primary key and Foreign Key)

```
CREATE TABLE students (
  student_id INT AUTO_INCREMENT PRIMARY KEY,
  student_no VARCHAR(20) NOT NULL UNIQUE,
  first_name VARCHAR(50) NOT NULL,
  last_name VARCHAR(50) NOT NULL,
  course VARCHAR(100),
  year_level INT CHECK (year_level BETWEEN 1 AND 5),
  email VARCHAR(100) UNIQUE,
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
) ENGINE=InnoDB;

CREATE TABLE assignments (
  assignment_id INT AUTO_INCREMENT PRIMARY KEY,
  assignment_code VARCHAR(20) NOT NULL UNIQUE,
  title VARCHAR(100) NOT NULL,
  description TEXT,
  due_date DATE,
  max_score DECIMAL(5,2) CHECK (max_score >= 0),
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
) ENGINE=InnoDB;

CREATE TABLE submissions (
  submission_id INT AUTO_INCREMENT PRIMARY KEY,
  student_id INT NOT NULL,
  assignment_id INT NOT NULL,
  submission_date DATETIME NOT NULL,
  file_path VARCHAR(255),
  score DECIMAL(5,2) CHECK (score >= 0),
  feedback TEXT,
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

  -- Relationships
  CONSTRAINT fk_sub_student FOREIGN KEY (student_id)
    REFERENCES students(student_id)
    ON DELETE CASCADE
    ON UPDATE CASCADE,

  CONSTRAINT fk_sub_assignment FOREIGN KEY (assignment_id)
    REFERENCES assignments(assignment_id)
    ON DELETE CASCADE
    ON UPDATE CASCADE,

  UNIQUE KEY uq_student_assignment (student_id, assignment_id)
) ENGINE=InnoDB;
```

2. Table Structure (Student, assignment and submission) - screenshot of the SQL statement and table Structure

A. Assignment Table

	assignment_id	assignment_code	title	description	due_date	max_score	created_at
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL

B. Student Table

	student_id	student_no	first_name	last_name	course	year_level	email	created_at
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

C. Submissions Table

	student_id	student_no	first_name	last_name	course	year_level	email	created_at
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

3. Relational schema from the EER tool of Workbench (pdf or jpg file)